

- 11.5.** Modify the subroutine written for problem 11.4 to eliminate all unnecessary multiplication and assignment. Test your subroutine against the result that was obtained in Example 11.7.1.
- 11.6.** Write a computer subroutine that uses the Euler-Rodrigues quaternion and the subroutine written for problem 11.1 to transform the components of an arbitrary vector from Earth-fixed coordinates to body-fixed coordinates. Test your subroutine against the result that was obtained in Example 11.7.1.
- 11.7.** Modify the subroutine written for problem 11.6 to eliminate all unnecessary multiplication and assignment. Test your subroutine against the result that was obtained in Example 11.7.1.
- 11.8.** Write a computer program that uses the fourth-order Runge-Kutta numerical integration method with the Euler-Rodrigues quaternion formulation to solve the problem in Example 11.9.1. Apply Eq. (11.10.7) to renormalize the quaternion after each full time step. Run this code for the case $V_v = 100$ ft/sec, $V_h = 200$ ft/sec, and $\omega = 2\pi$ sec⁻¹. Compare your numerical solution with the analytical solution obtained in Example 11.9.1.
- 11.9.** Modify the computer program written for problem 11.8 to eliminate that part of the code, which is used to renormalize the quaternion after each full time step. Using this modified code, repeat the computations made in problem 11.8. Compare your numerical solution with the analytical solution obtained in Example 11.9.1.
- 11.10.** Write a computer program to repeat the computations made in Example 11.11.2. Use the fourth-order Runge-Kutta numerical method to integrate the Euler-Rodrigues quaternion formulation. Apply Eq. (11.10.7) to renormalize the quaternion after each full time step. Test your program against the numerical results that were presented in Example 11.11.2. To test your program even further, use an initial condition that has the arrow rotated 90 degrees about the x_b -axis. Also try an initial condition that has a heading of 225 degrees. You should be able to deduce how these solutions will differ from the one presented in Example 11.11.2.
- 11.11.** Write a computer program to repeat the computations made in Example 11.11.3. Use the fourth-order Runge-Kutta numerical method to integrate the Euler-Rodrigues quaternion formulation. Apply Eq. (11.10.7) to renormalize the quaternion after each full time step. Test your program against the numerical results that were presented in Example 11.11.3. To test your program even further, use an initial condition that has the arrow rotated 90 degrees about the x_b -axis. Also try an initial condition that has a heading of 225 degrees. You should be able to deduce how these solutions will differ from the one presented in Example 11.11.3.